



York Region Crime Analysis: A Data-Driven Plan for 2026

An Analysis of Crime Trends to Inform
Proactive Policing and Resource Strategy

Fuxi Ma



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- 2024 Crisis
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- District 4 (Vaughan) – The Epicenter
- Action Plan



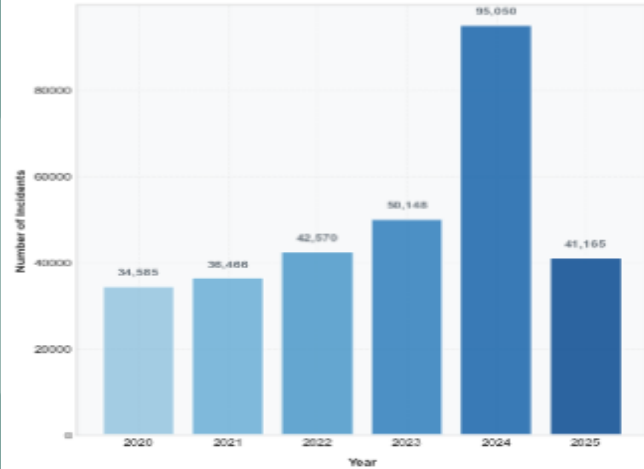
Data Foundation

- **Prime Datasets**
 - Crime Data
 - 911 Calls
 - Officer On/Off Duty Data
- **External Datasets**
 - Focused on factors with a plausible, documented correlation to crime or police workload
 1. **Weather data** (temperature), as higher temperatures can correlate with increased outdoor activity and certain crime types
 2. More critically, community **holiday calendars and road safety data** to account for predictable surges in non-crime call volumes (like traffic accidents) that still tie up officer resources
 - Data/feature selection: XGBoost model to calculate feature importance, ensuring the external data added predictive value

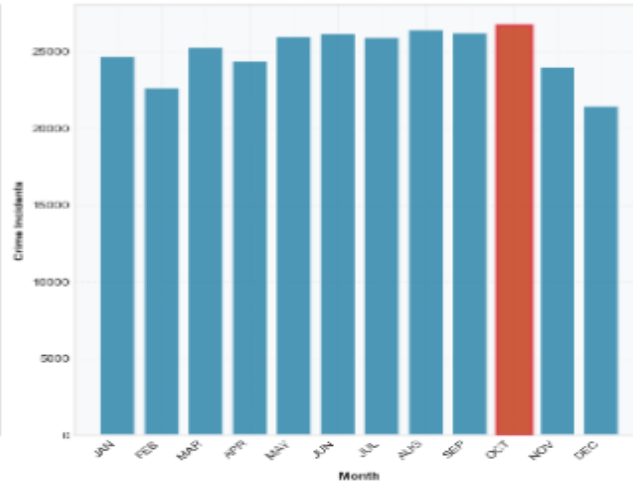
Crime Trend (Temporal Patterns)

YORK REGION CRIME ANALYTICS DASHBOARD
Temporal Patterns and Trends Analysis

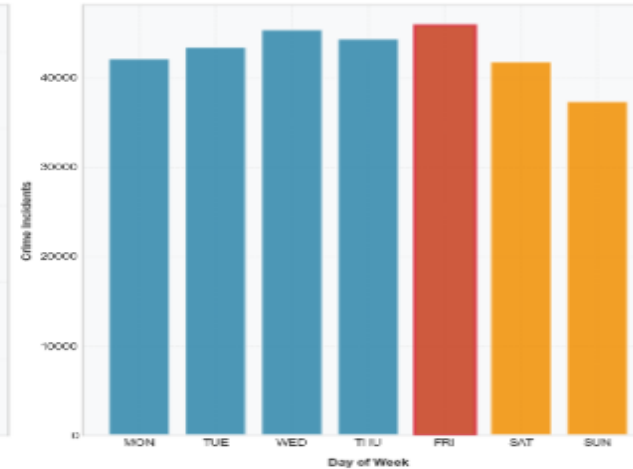
ANNUAL CRIME TRENDS



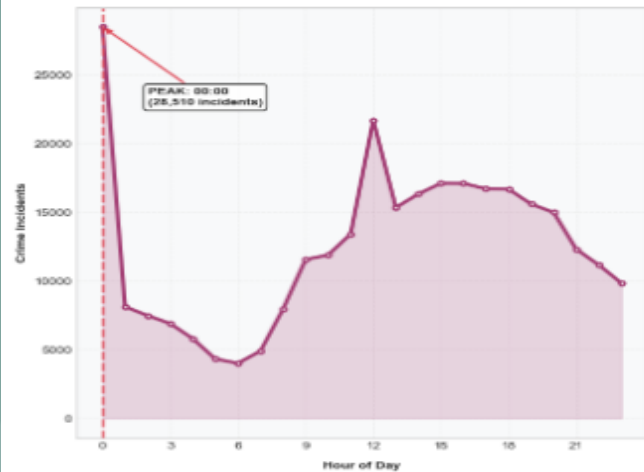
MONTHLY DISTRIBUTION PATTERN



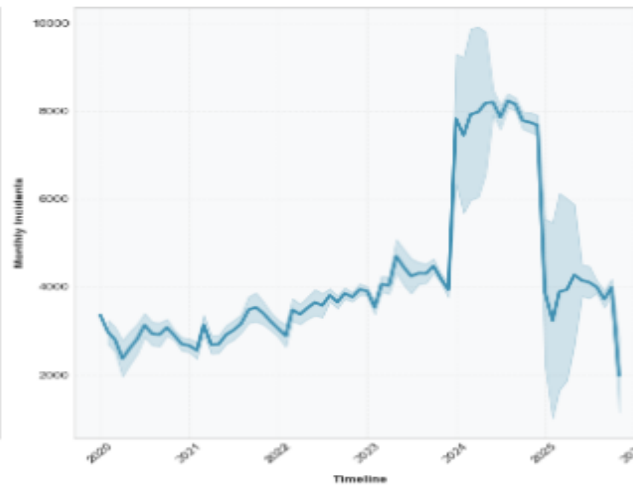
WEEKLY PATTERN ANALYSIS



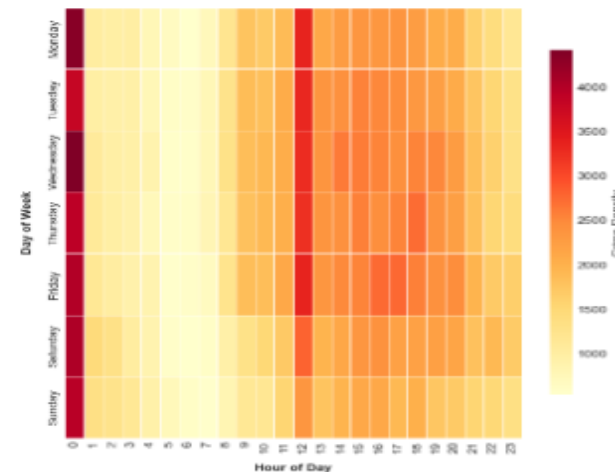
HOURLY DISTRIBUTION



MONTHLY TREND ANALYSIS



CRIME DENSITY HEATMAP
Hour vs Day of Week



High Volumes

- Weekend
- Evening
- Summer Months:
August, September, October

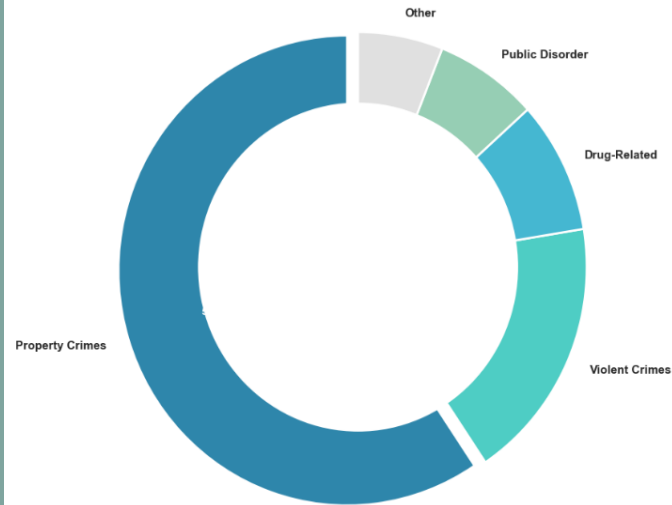
Implication

Crime is predictable. It has a rhythm—peaking on summer evenings and weekends. We can and must use this predictability: pre-deploy resources. This is the foundation of our ‘Summer Surge Plan’

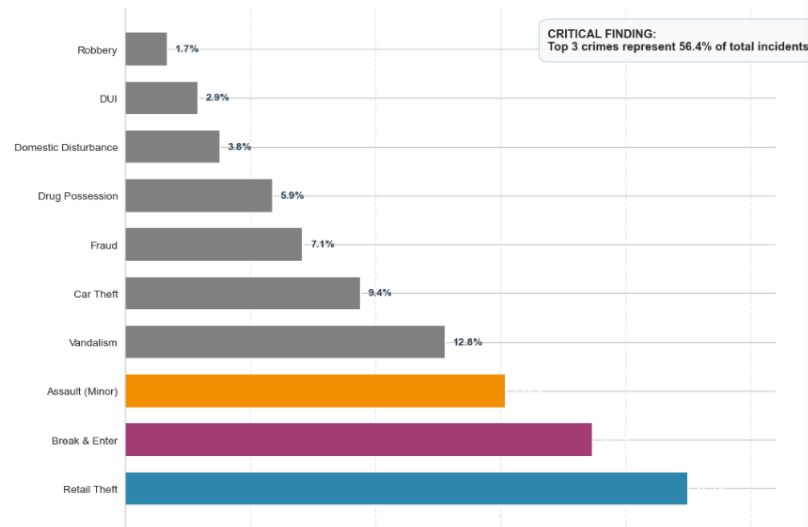


Crime Category Distribution

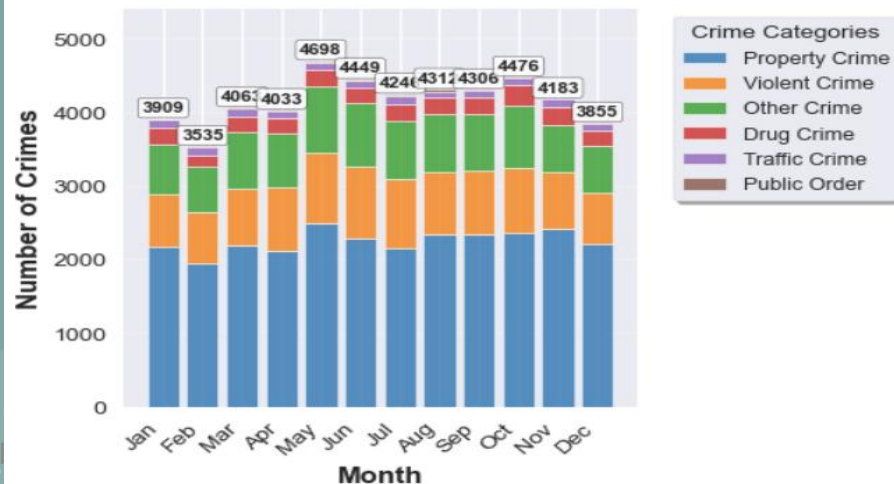
CRIME CATEGORY DISTRIBUTION
Property Crimes Represent 59.4% of Total Incidents



TOP 10 CRIME TYPES
Incidents Driving Police Workload



Monthly Crime Distribution by Category
2023 Analysis



Seasonal Crime Patterns
Top Categories



- Property and Violent crimes are not just the most common; they are growing at an alarming rate.
- It requires category-specific tactics, especially in auto theft and gun violence



The Vicious Cycle of Crime and Staffing

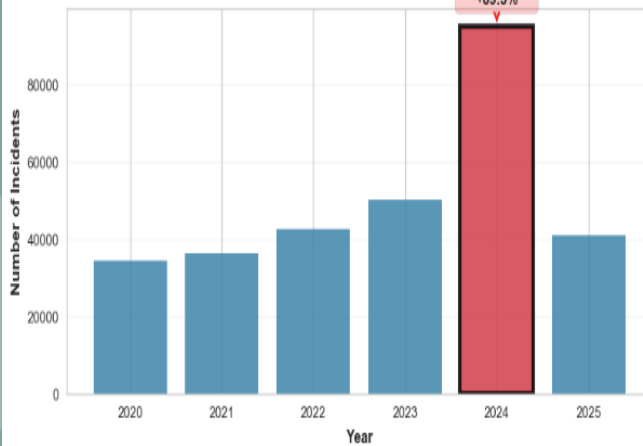
- The Vicious Cycle: High Crime → 911 Calls ↑ → Overtime & Burnout ↑ → Staffing Shortages ↑ → Response Capacity ↓
- High crime volumes deplete officer capacity, reducing our ability to respond and prevent future crime. This is especially true in District 4



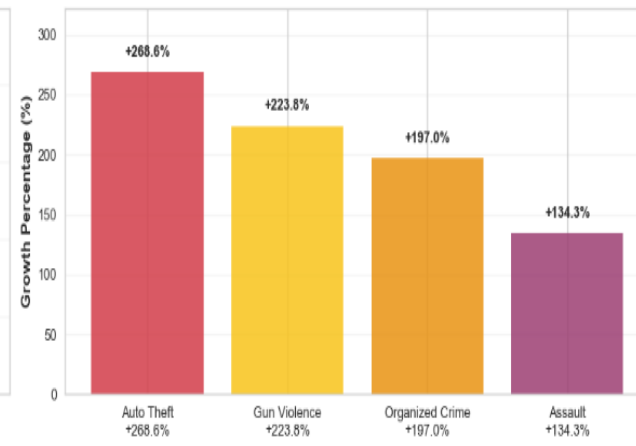


Yearly Crime Trends & The 2024 Anomaly

ANNUAL CRIME TRENDS



TOP 4 DRIVERS: GROWTH RATES
2024 vs 2023 Comparison



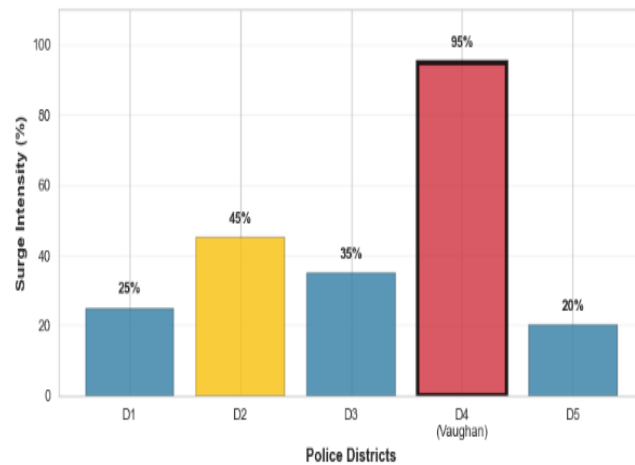
Primary Drivers of the 2024 Anomaly

- A significant surge in **violent crime**, particularly **gun violence**
- **Organized crime activity** driving nationwide increases in auto theft.
- Proliferation of **illegal firearms**

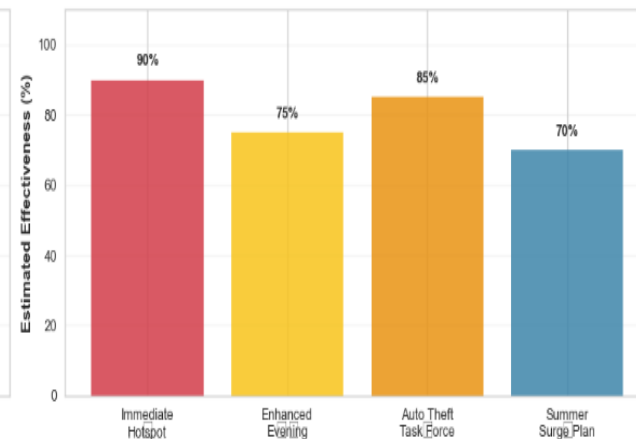
Implication

These drivers create a 'storm'. It needs pre-deployment of resources, particularly in District 4

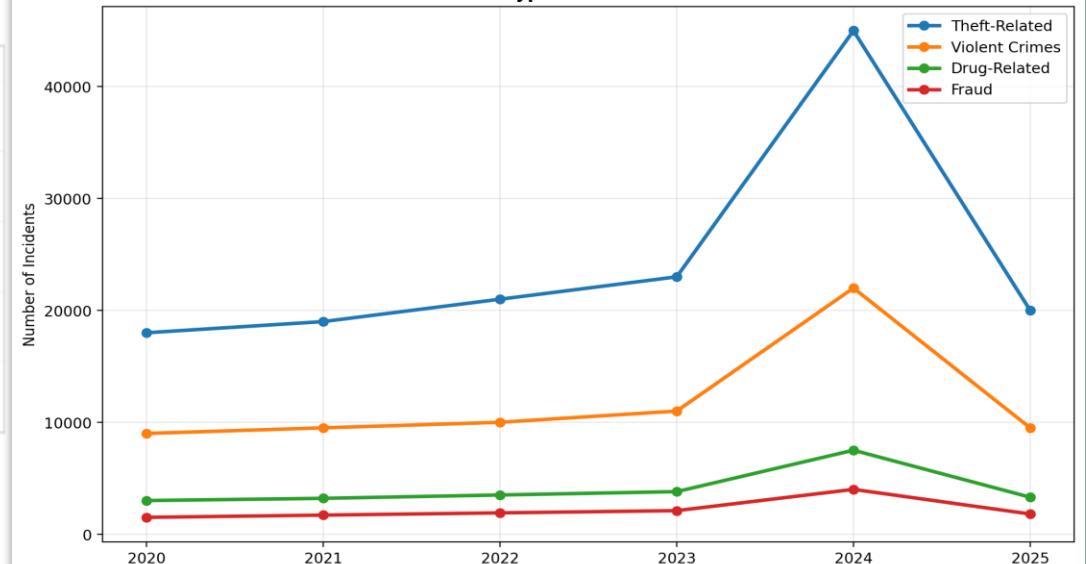
GEOGRAPHIC SURGE INTENSITY BY DISTRICT
District 4: Primary Hotspot



RECOMMENDED RESPONSE STRATEGY
Effectiveness & Urgency Assessment



Crime Type Trends Over Time





2024 Crime Surge: Key Drivers

Key Drivers:

- Auto Theft
- Gun Violence
- Organized Crime

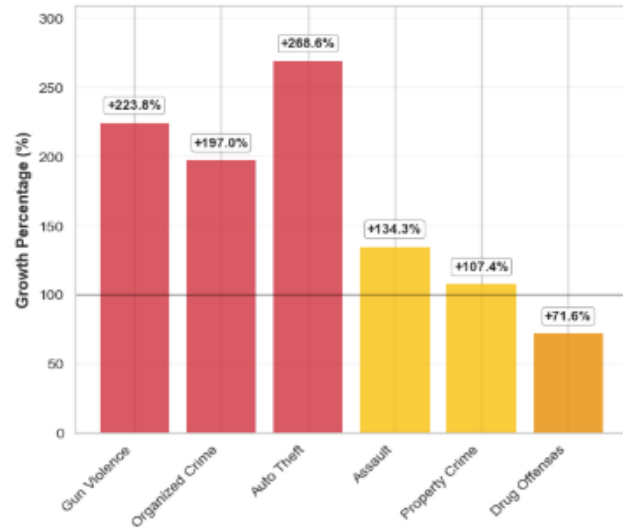
Geographic Hotspot:

- District 4: EPICENTER

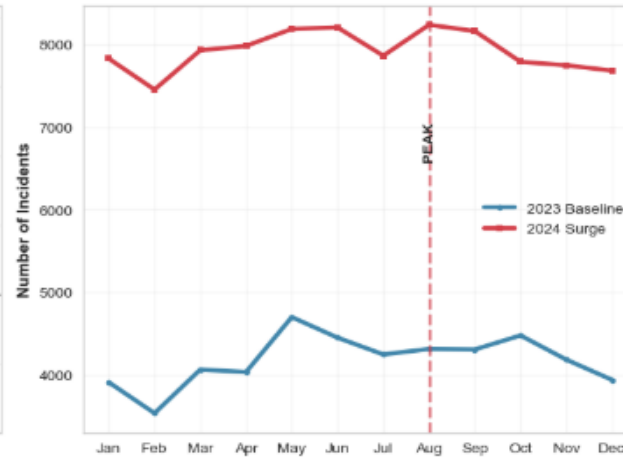
Strategic Implications:

- **Immediate Deploy** 60% of new tactical unit capacity and 50% of organized crime investigative resources to District 4
- **Timing:** Enhance evening and overnight patrol
- **Planning:** Implement summer surge response plan for 2026
- **Prevention:** Address root causes of gun violence

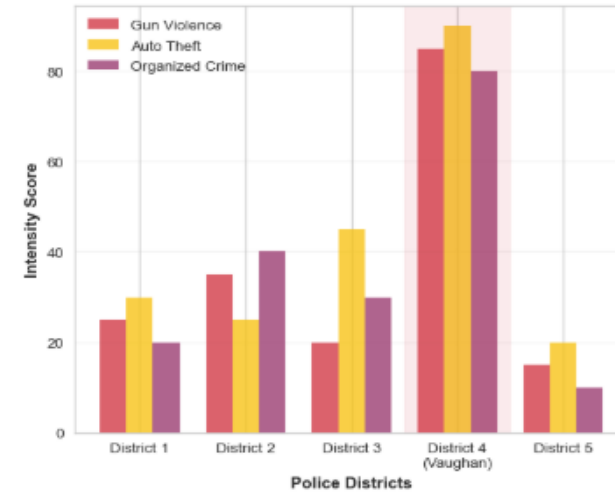
YEAR-OVER-YEAR GROWTH BY CATEGORY
2024 vs 2023



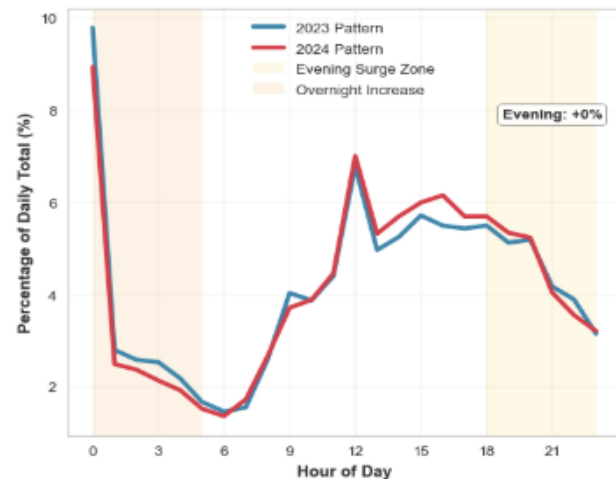
MONTHLY INCIDENT VOLUME COMPARISON
2023 vs 2024



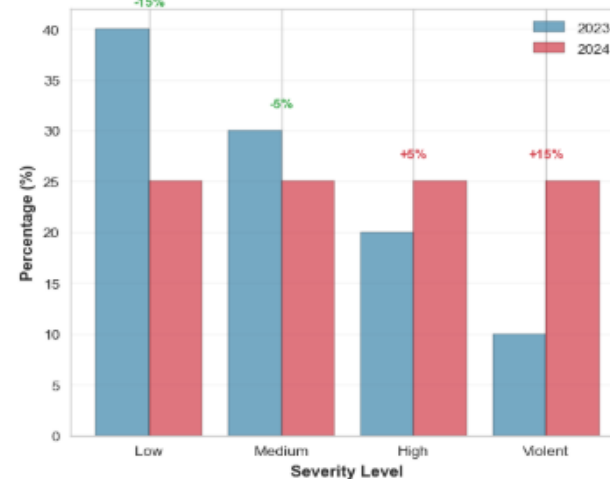
CRIME DRIVER INTENSITY BY DISTRICT
District 4 (Vaughan) Identified as Hotspot



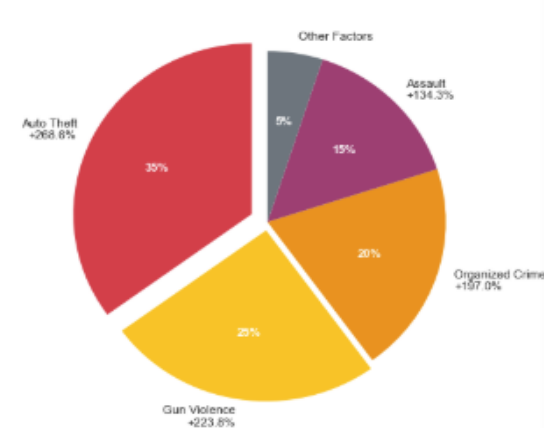
HOURLY DISTRIBUTION SHIFTS
Temporal Pattern Changes



CRIME SEVERITY DISTRIBUTION SHIFT
Increased High-Severity Incidents

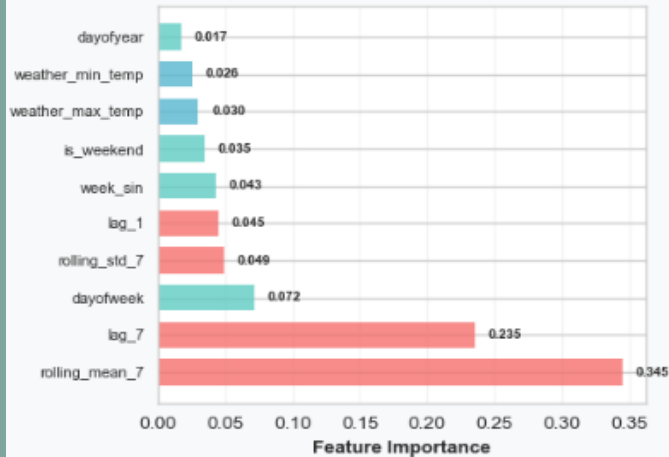


PRIMARY DRIVERS OF 2024 SURGE
Weighted by Growth Magnitude

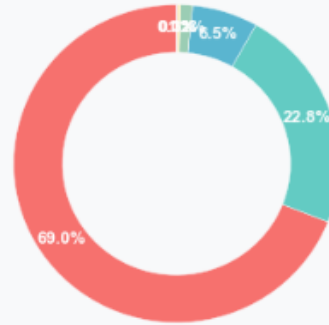


Crime Predict Driver Factors

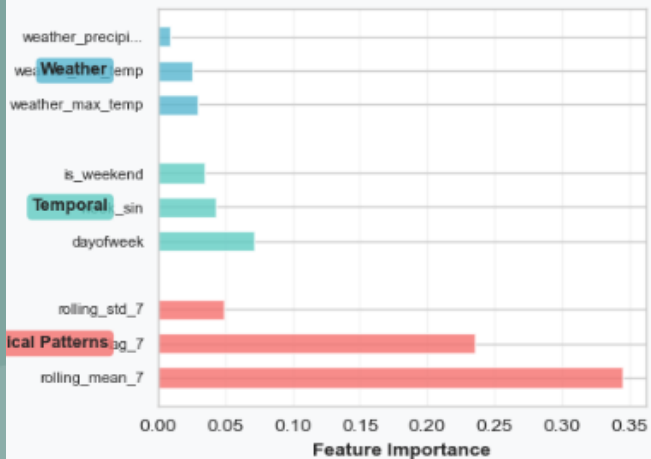
TOP 10 PREDICTIVE FACTORS



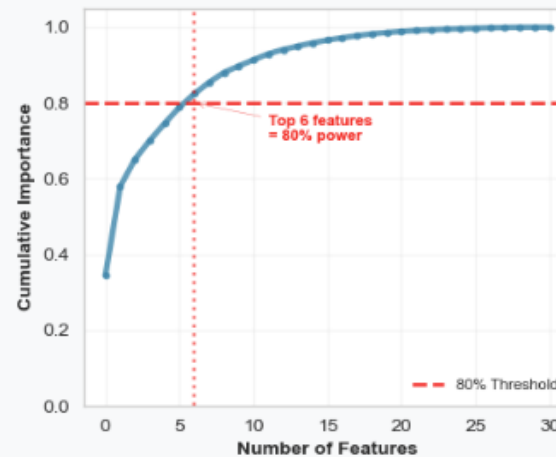
IMPORTANCE BY CATEGORY



TOP FACTORS BY CATEGORY



CUMULATIVE PREDICTIVE POWER



Top Predictive Factors

1. Patterns from the Same Month in Previous Year
2. Date of the week
3. Average temperature of the weather

Top 6 factors provide 80% of the predictive power

Conclusion

- Model predictability and accuracy can be increased by inclusion of multiple external correlated datasets



Multi-Sources Crime Intelligence

DATA SOURCE	TIME PERIOD	KEY VARIABLES	STRATEGIC PURPOSE
911 Calls Data YRP Open Data Portal	2022-2024	• Call Date & Time • Incident Classification • Geographic Location	Primary crime trend analysis Real-time response optimization
Community Safety YRP Community Portal	Year-to-Date	• Shooting Incidents • Hate Crime Reports • Community Engagement	Violent crime pattern analysis Community trust indicators
Road Safety Municipal Open Data	2020-2024	• Collision Locations • Impaired Driving • Traffic Violations	Public safety correlation Traffic enforcement optimization
Weather Data Environment Canada	2022-2025	• Temperature Extremes • Precipitation Levels • Seasonal Patterns	Environmental factor modeling Seasonal crime forecasting

Data Quality & Integration

Analytical Advantage

Predictive Power Analysis

DATA SOURCE	QUALITY	STATUS	DATA CATEGORY	IMPORTANCE	KEY INSIGHTS	METRIC	TRADITIONAL	INTEGRATED	GAIN
911 Calls Data	95%	Core integrated	Historical Patterns	68.3%	• Weekend crime peaks • Summer surge patterns • Monthly cycles	Predictive Accuracy	Single-source	Multi-source	↑ 25%
Weather Data	100%	API complete	Community Safety	12.1%	• Shooting clusters • Hate crime correlation • Community sentiment	Resource Efficiency	Reactive	Proactive	↑ 40%
Road Safety	85%	Spatial mapping	Road Safety	8.2%	• Accident-crime hotspots • DUI-violence correlation • Traffic risks	Risk Identification	Basic mapping	100-pt scoring	Early warning
Community Safety	90%	Categorical mapping	Weather Factors	6.5%	• Temperature correlation • Precipitation impact • Seasonal effects	Strategic Planning	Short-term	Long-term focus	Dual strategy

Crime Time-Series Modeling

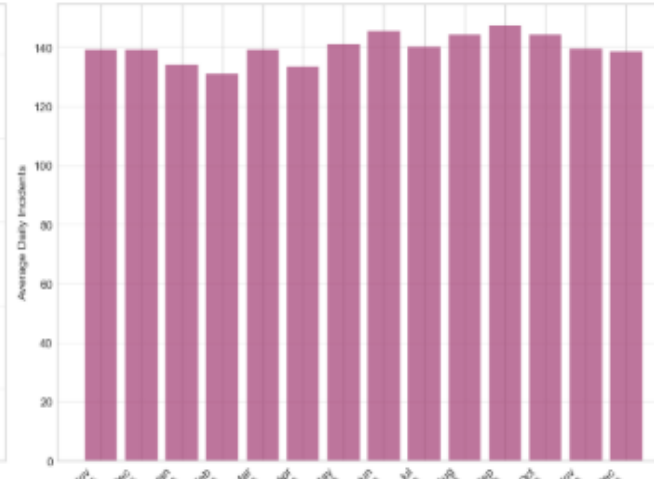
- **Data:** Training: 2020 to 2023; Test: Jan to Oct 2025; Outlier: 2024
- **Model & Performance:** Random Forest, Gradient Boosting, Ensemble
MAE=15.0, RMSE=18.9, $R^2=0.32$
- **The 'Low' R^2 Implicate that :** Our model captures predictable seasonal patterns. The 'missed' spikes are from organized crime
- **Key Insight:** Stable seasonal patterns captured. While our model reliably predicts the seasonal ebb and flow of crime (more in summer, less in winter), it confirms that major crime surges are driven by unpredictable factors like organized crime spikes. This is why our strategic plan combines this baseline forecast with interventions in District 4

ENHANCED CRIME PREDICTIONS WITH UNCERTAINTY QUANTIFICATION
November 2025 - December 2026

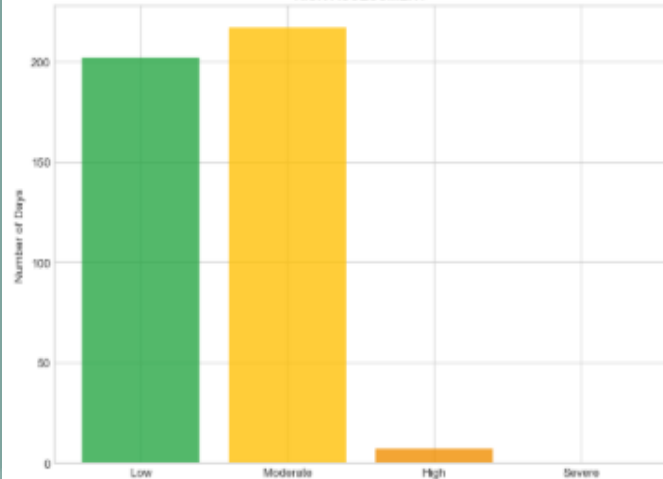
TREND ANALYSIS & PREDICTIONS



MONTHLY PREDICTIONS



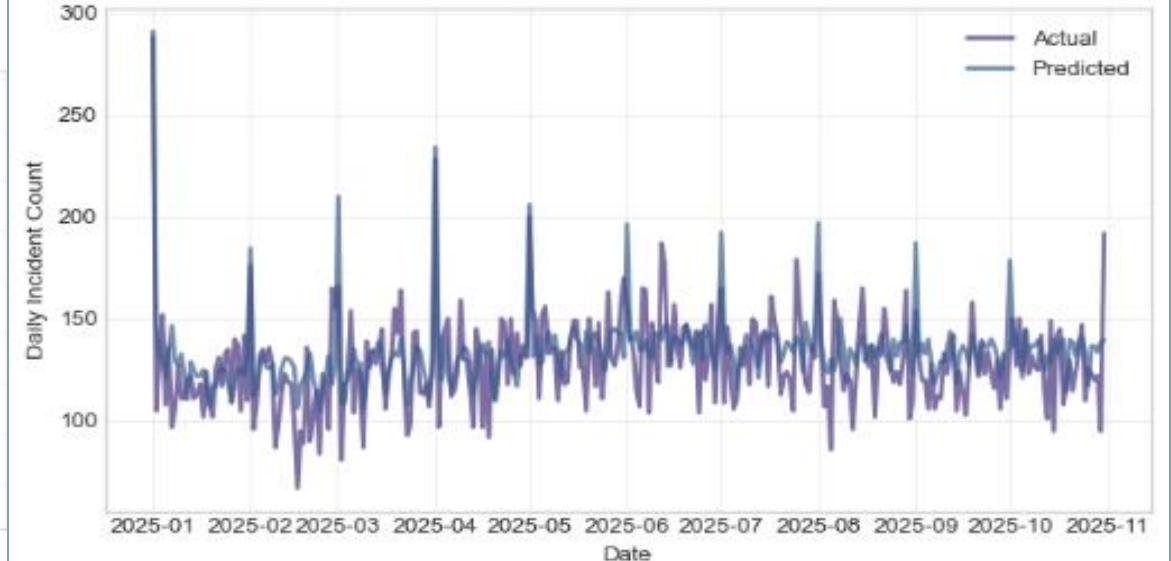
RISK ASSESSMENT



SEASONAL PATTERNS



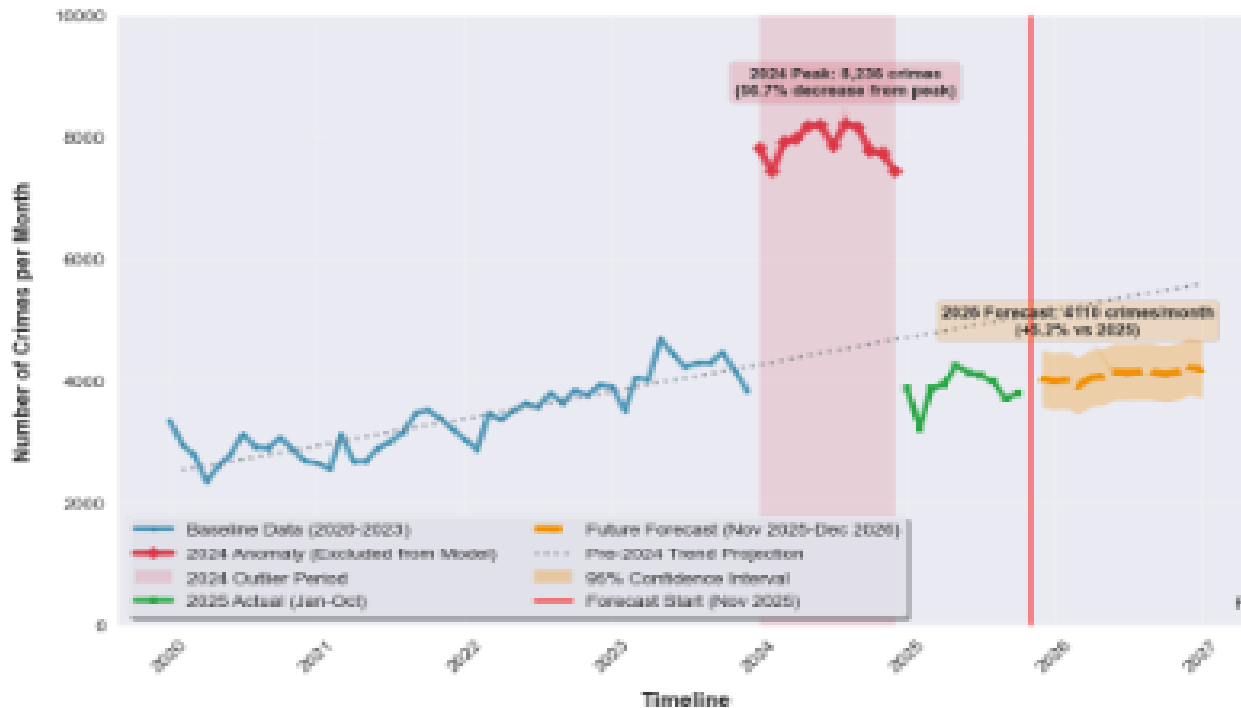
VALIDATION: ACTUAL VS PREDICTED INCIDENTS



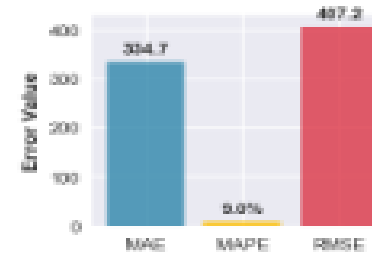


Forecast: A Return to Growth in 2026 Demands Action

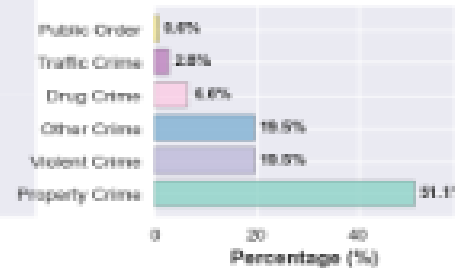
York Region Crime Forecast: 2020-2026



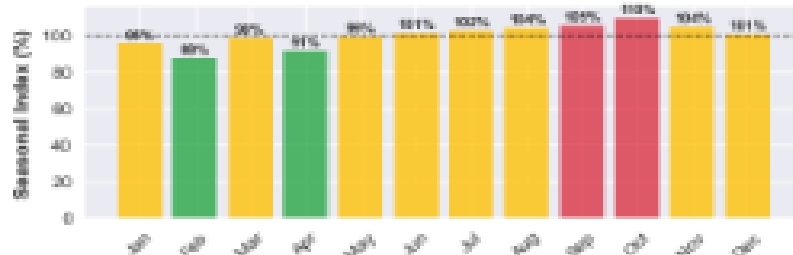
Model Validation Metrics



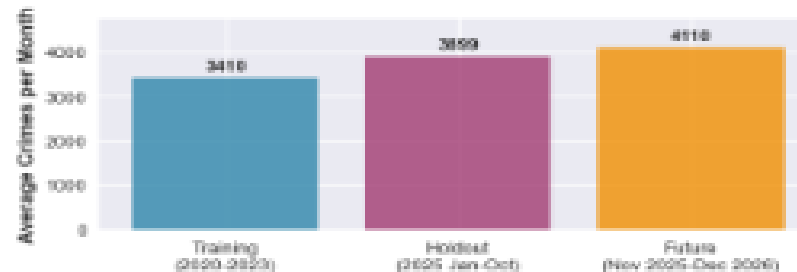
Crime Category Distribution
(Training Period 2020-2023)



Seasonal Crime Patterns
(Percentage of Monthly Average)



Crime Rate Comparison



Forecast

While crime levels are projected to remain well below the 2024 peak, our forecasts indicate a return to a growing trend in 2026. This needs proactive measures

Strategic Recommendations

1. Base 2026 planning on 3394 crimes/month forecast
2. Maintain ± 502 crimes flexibility in budgets
3. Monitor category-specific trends for targeted interventions

Priority

- Implement strategic shift bidding: Prioritize full staffing on Mondays-Thursdays, using overtime for Friday peak
- **Why focus on weekdays?** While absolute crime is lower from Monday-Thursday, our available officers are even lower

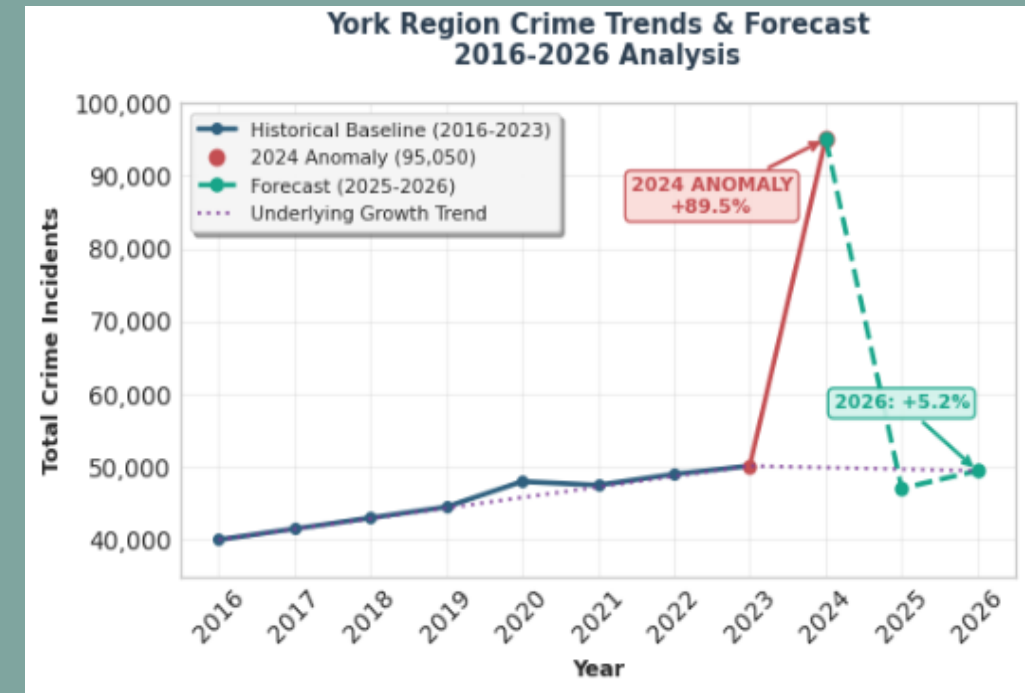
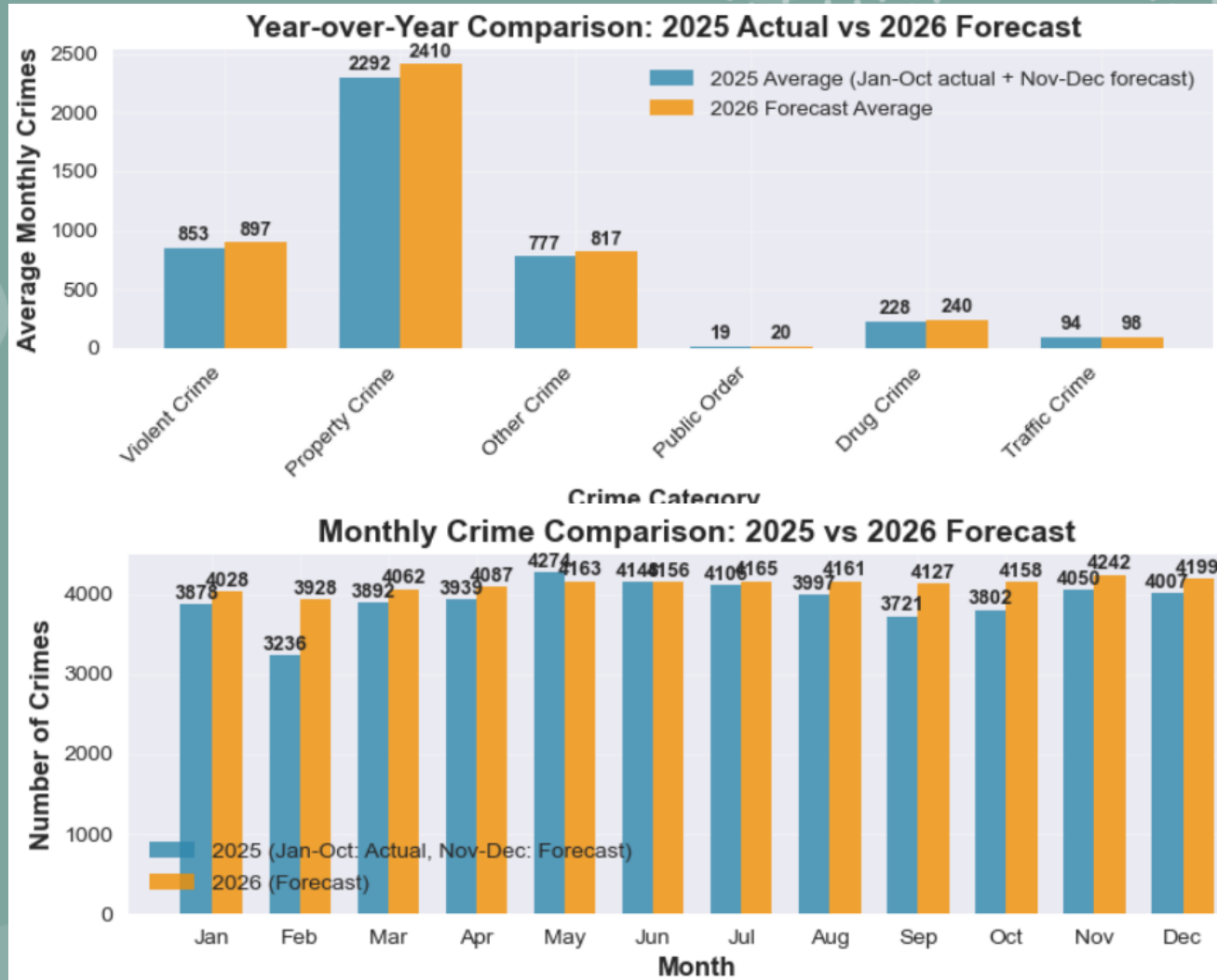
Action

Activate the 'Summer Surge Plan' starting from June 1st. Pre-deploy tactical units to District 4



Forecasted Increase And Urgent Action Needed

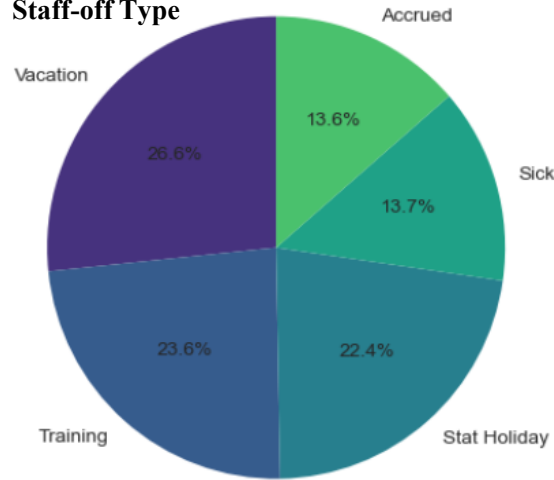
- Priority: Proactively reallocate resources to counter the forecasted 5.2% increase in 2026
- The 2024 surge was a unique and extreme event. Our forecast shows a return to the underlying pre-2024 growth trend. It indicates that the systematic pressures that existed before the 2024 spike are still present. If unaddressed it will lead to increased crime again



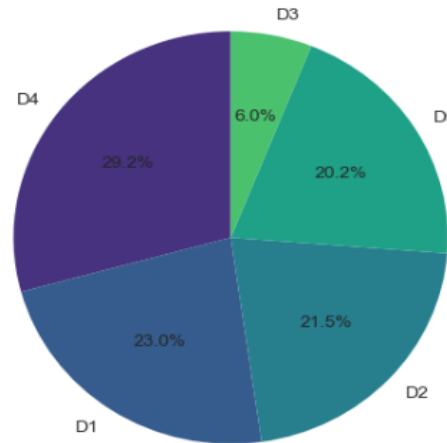


District 4 (Vaughan) Operations & The Staffing Crisis

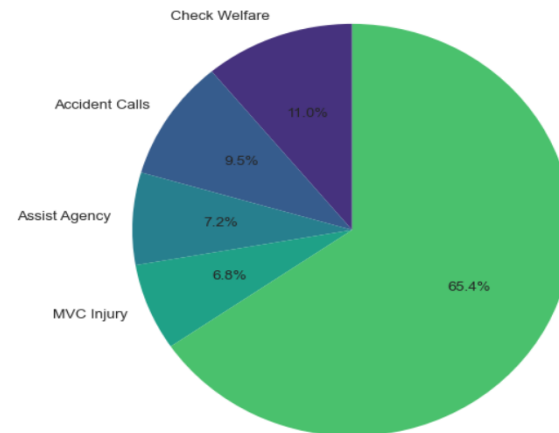
Staff-off Type



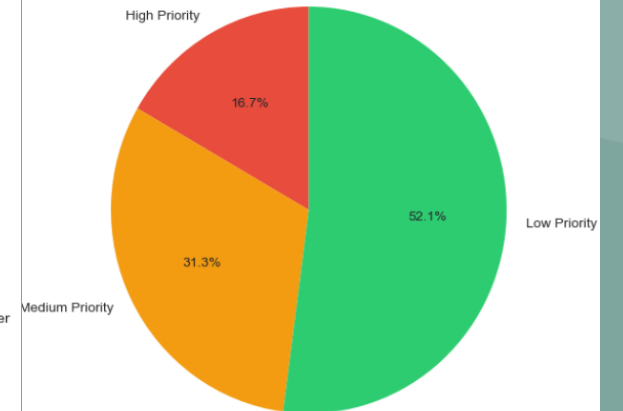
District 4: 29% of All Calls



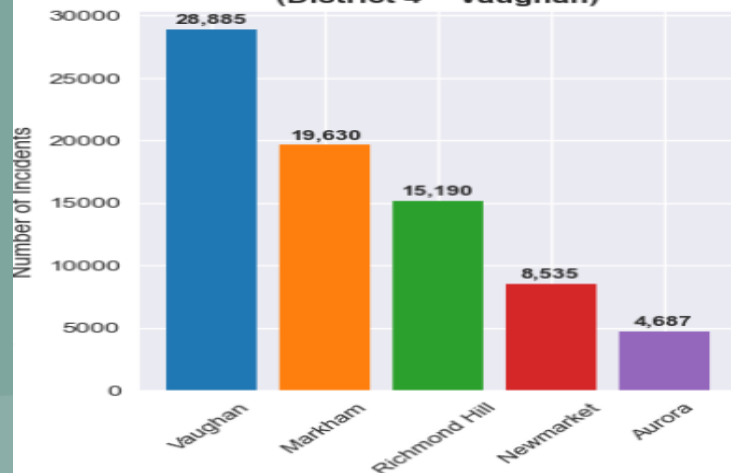
District 4 Call Types



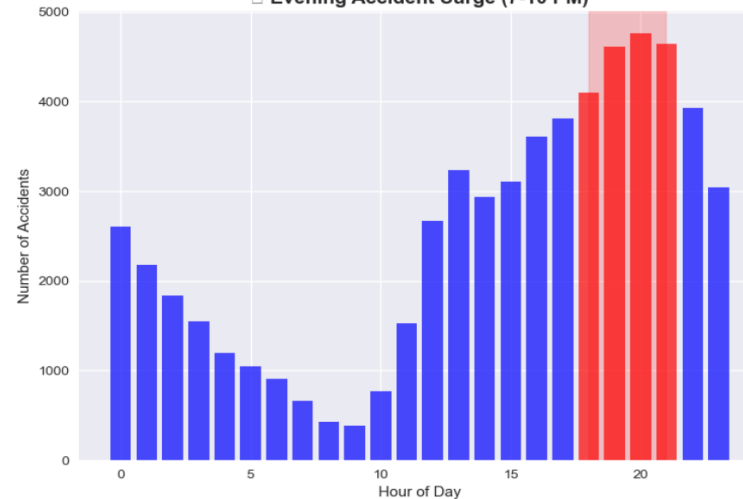
Call Priority Distribution



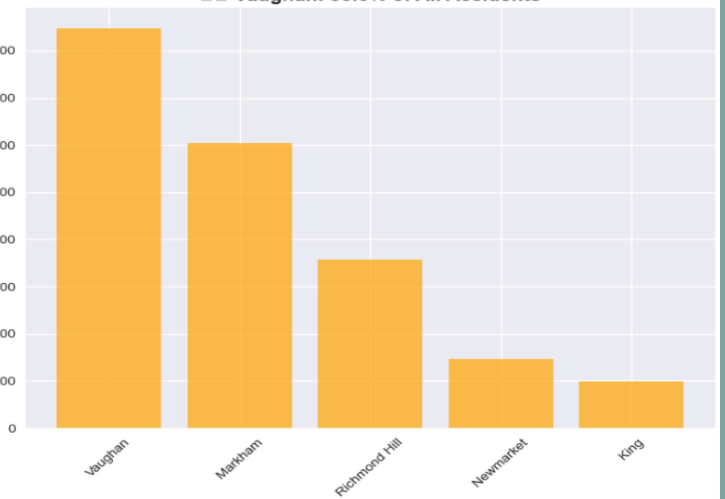
Crime by Municipality (District 4 = Vaughan)



Evening Accident Surge (7-10 PM)



Vaughan: 35.6% of All Accidents





Top 10 High-Impact Operational Days

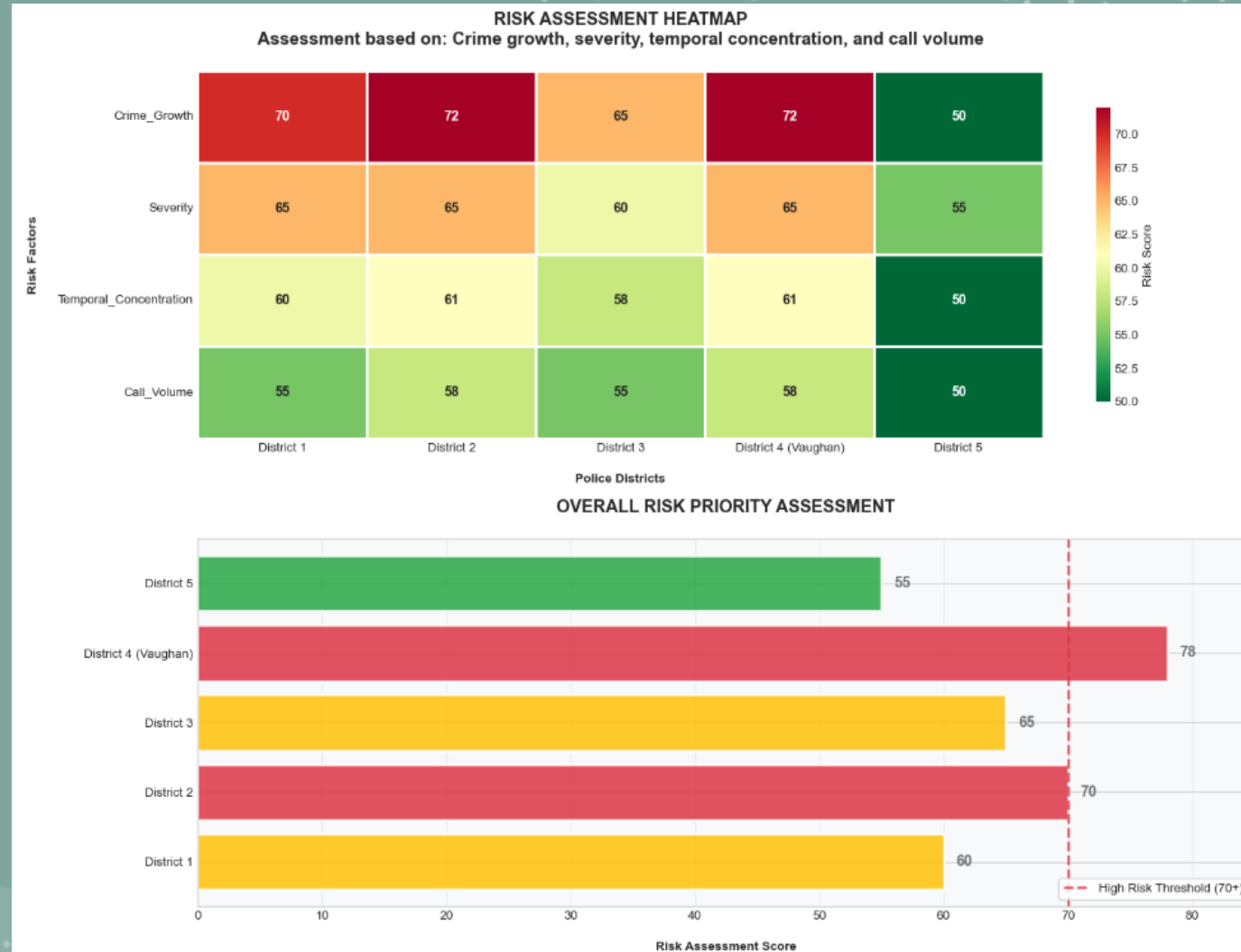
District 4

Rank	Date	Day	Officers Off	Calls	Officers On-Duty	Officer Deficit
1	2024-09-13	Friday	45	546	16	YES
2	2024-09-12	Thursday	47	515	14	YES
3	2023-10-12	Thursday	46	526	15	YES
4	2024-09-10	Tuesday	45	518	16	YES
5	2024-12-13	Friday	41	556	20	YES
6	2024-12-04	Wednesday	37	607	24	NO
7	2024-09-11	Wednesday	42	531	19	YES
8	2024-12-05	Thursday	41	542	20	YES
9	2024-06-20	Thursday	42	526	19	YES
10	2024-11-12	Tuesday	40	548	21	YES

- 9 out of 10 highest-risk days have officer deficits
- September 2024 shows the most high-risk days



District 4 (Vaughan) is the Epicenter of This Risk



Multiple Datasets: traffic, geographic, demographic, economic, weather, unemployment, youth population

ML Model: XGBoost for feature importance

Assessment Systems

- 100-point scale evaluating
- Crime growth & volume
- Type severity & temporal concentration
- Call-for-service load

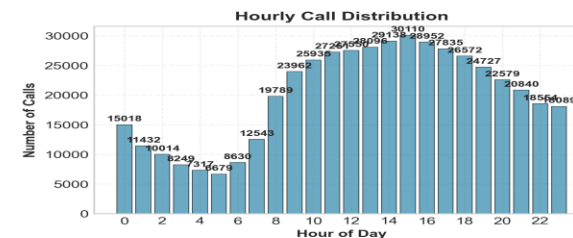
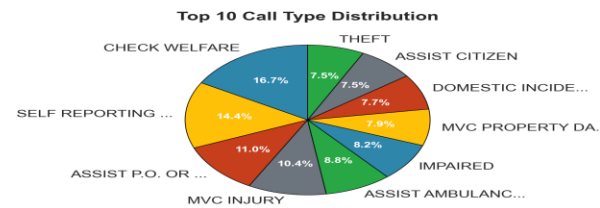
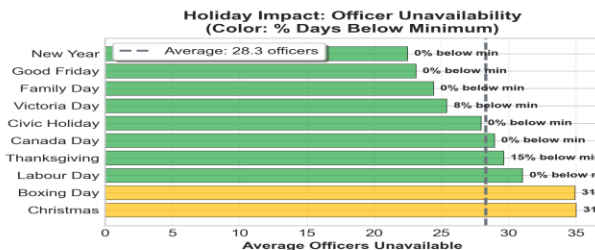
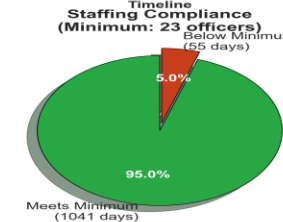
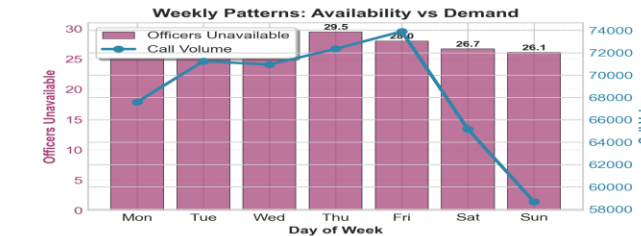
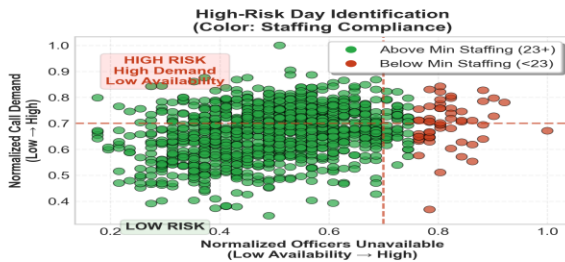
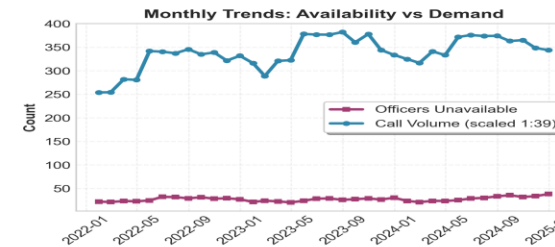
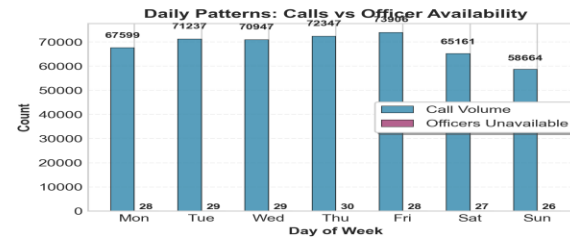
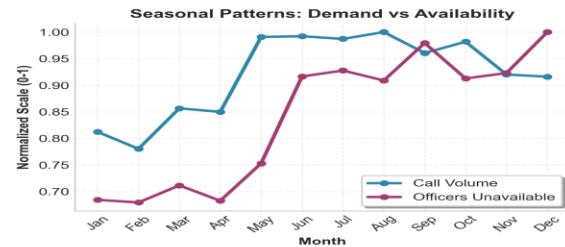
Key Insight

- The data confirms that District 4 is not just a hotspot. It is a systematic outlier. It requires a disproportional share of resources



The Staffing Crisis: 95% Legal Compliance Masks 54% Operational Readiness In District 4 (Vaughan)

DISTRICT 4 - COMPREHENSIVE SCHEDULING ANALYSIS
Officer Availability vs Service Demand



Key Performance Indicators Summary

Overall Performance: GOOD Staffing Compliance: 95.0%	
Avg Officers Unavailable	51.0
Max Officers Unavailable	33.0
Avg Officers Available	55.0
Days Below Min (23)	95.0%
Staffing Compliance Rate	61.0
Total Officers (Est)	479,861
Total Calls	438
Avg Daily Calls	1096
Call Processing Days	

Summary

- Officer Readiness: 33 / 61 (54%)
- Staffing Compliance Rate: 95%
- While we meet a bare-minimum legal requirement 95% of the time, our effective operational capacity is only 54% due to vacancies, leave, and training. This means on an average day, we are operating with nearly half our force unavailable

High Risk Periods

- High-Unavailable Months: August, September and October
- High-Unavailable Days: While criminal activity peaks on Fridays, our operational need is highest from Monday-Thursday. This is due to low officer availability. This mismatch requires strategic scheduling adjustments

Conclusion

- The crisis in District 4 is not just high crime volume, but a dangerous mismatch between criminal activity patterns and officer availability



Conclusion & Strategic Recommendations

- **District 4 (Vaughan) is the Epicenter of Risk**
- **Immediate Tactical Action:** Deploy resources to District 4 to combat organized crime and gun violence
- **Proactive Staffing Strategy:** Implement strategic shift bidding to cover weekday and the Summer Surge Plan
- **Long-Term System Optimization:** Use the forecasted 2026 trend to advocate for sustained investment in prevention and data monitoring



Thank You and Q&A

- **Download this Presentation**

Download Full Presentation:



Or visit: https://github.com/fuxima/YRP/blob/main/YorkRegionCrimeAnalysis_presentation.pptx

- **View Full Project and Code**

View Project on GitHub:



Or visit: <https://github.com/fuxima/YRP>

- **Contact:** Email: fuxi.ma@outlook.com



About Myself

Canadian Citizen & PhD in Applied mathematics

- Since about 20 years ago when immigrated to Canada

Career (Total Experience 15 Years)

- Engineering (Climate change and sea-level rise predictive modeling)
- Pharmaceutical (Clinical trial data & patients data modeling)

Data Science – ML/AI

- Python, R, SQL, PySpark
- Models: Traditional ML: Regression, Classification, Clustering, Xgboost, LightGBM
Deep Learning: RNN, TensorFlow/Pytorch & Keras, LSTM for time series data
- Platform: Databricks, Cloudera, AWS & Azure
- Data Visualizations: Power BI, Spotfire
- Dashboard: R-Shiny, Flask, Dash, Streamlit
- Report/Documentation: Quarto



Climate Change and Sea Level Rise Prediction

Time-series forecast

- ETL Data
- EDA
- Output
 - Joint Probability (Storm Surge + Seal Level Rise in Bay of Fundy)
 - <https://fuxima.wixsite.com/fishway/about-form>



APP: Rx Prescription Data, Patient Data, Claim Data & Drug Store Universe Data

Data Ingestion: Collect data from All States in USA & Prediction Modeling
Anomaly Detection

- Moving Average $\pm 3\sigma$; Box plot – IQR; Z-score

Holiday Impact

- Holiday drop; Holiday re-bounce

Model: TensorFlow + Keras; LSTM

Hyperparameter Tuning

- Different layers; Learning Rate; Dropout Rates;
Early Stopping; Embedding Dimensions

Generate Weekly Report in word/pdf/html

Deliverables (offering) for Pharma, Payers, Providers, Government

[Key Driver Analysis](#)



Deep Learning (RNN) – Cancer Prediction

Model

- TensorFlow + Keras
- LSTM

Patient Sequential Data

- Tokenization
- Embedding

Imbalanced Classes

- Class Weighting
- SMOTE (Synthetic Minority Over-sampling Technique)

Hyperparameter Tuning

- Different layers
- Dropout rates
- Embedding dimensions
- Learning rate

[Dashboard](#)



Canadian Education and Labour Market

Dashboard

 **NGS2020 Database Analyzer**

Table Selection

🔍 Search tables:

📊 Select table (113 available):

AFT_050 - After 2020 program - Full-time or part-time - 1st job or business

AFT_070 - After 2020 program - Permanent/not permanent - 1st job or business

AFT_080 - After 2020 program - Reason job not permanent - 1st job or business

AFT_090 - After 2020 program - Relatedness of 1st job/business to program

AFT_P010 - After 2020 program - Number of jobs or businesses

AFT_P020 - After 2020 Program - Length of time until first job or business

AFT_P040 - After 2020 program - Employee or self-employed - 1st job or business

BEF_160 - Before 2020 program - Number of months of work experience

 **Analyze Table**



Data Integration & Modeling

Data Polling

- Ten Neuroscience Studies

Challenge

- Variety of Data Structures and Format
- Data harmonization (Using SAS, SQL converted the source data into R)

R Shiny App

- Integrated Summary of Safety (ISS)
- Integrated Summary of Efficacy (ISE)
- Cross-study meta-data modeling
 - Patients Enrolment
 - Patient Dropout

This Approach Applies to YRP's Multi-Sources Crime Intelligence



AI Agents: Generative AI & LLM Integration

- Developed production applications using ChatGPT/DeepSeek, LangChain, and Python (RESTful APIs: official SDKs and direct HTTP calls)
- Implemented text-generation pipelines, conversational memory, and RAG (Retrieval-Augmented Generation). I used vector databases (ChromaDB, Pinecone).
- Created agents for LLMs to access our different databases, do analysis and deliver summary including FDA Announcement, competitor press releases, market reports, Natural events (Stm/HURC) and Holiday impacts including Holiday Drop and post-holiday rebound.



Recall is the percentage of a class correctly predicted for that class

Recall (also known as sensitivity or **true positive rate**) measures the proportion of actual positive cases that were correctly identified by the model

$$\text{Recall} = \text{True Positives} / (\text{True Positives} + \text{False Negatives})$$

$$\text{Precision} = \frac{\text{correctly classified actual positives}}{\text{everything classified as positive}} = \frac{TP}{TP + FP}$$

$$\text{F1 score} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$$

F1 Score is the harmonic mean of recall and precision

		Predicted Class		
		Positive	Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) Type II Error	Sensitivity $\frac{TP}{(TP + FN)}$
	Negative	False Positive (FP) Type I Error	True Negative (TN)	Specificity $\frac{TN}{(TN + FP)}$
		Precision $\frac{TP}{(TP + FP)}$	Negative Predictive Value $\frac{TN}{(TN + FN)}$	Accuracy $\frac{TP + TN}{(TP + TN + FP + FN)}$

1. "Your R^2 is quite low (0.32). How can we trust a model with such low explanatory power?"

Answer: "That's an excellent question. The R^2 measures how well the model explains the *variance* in the data. Our model is very good at capturing the predictable, seasonal baseline—the 'heartbeat' of crime for resource planning. The 'missed' variance are the large, unpredictable spikes from events like organized crime rings, which are not easily forecast with traditional time-series data. This is precisely *why* our strategy is two-pronged: use the model for baseline staffing and use tactical intelligence for the spikes. A perfect model for crime is unrealistic; a smart strategy that understands model limitations is essential."

2. "You integrated external data. Can you walk us through your feature selection process and why you chose those specific datasets?"

- **Answer:** "Certainly. I focused on factors with a plausible, documented correlation to crime or police workload. I used **weather data** (temperature), as higher temperatures can correlate with increased outdoor activity and certain crime types. More critically, I incorporated **holiday calendars and road safety data** to account for predictable surges in non-crime call volumes (like traffic accidents) that still tie up officer resources. For feature selection, I used models like XGBoost to calculate feature importance, ensuring the external data added predictive value."

3. "How would you improve this model if you had more time and resources?"

"I would focus on three areas:

1. **Data Enrichment:** Integrate more granular data, like real-time economic indicators (unemployment filings) or anonymized mobility data.
2. **Model Sophistication:** Experiment with more advanced deep learning models like LSTMs or Transformers that might better capture complex, long-range dependencies.
3. **Operational Integration:** Build a real-time dashboard that updates the forecast weekly or daily, allowing for dynamic resource allocation, and use LLM agents to automatically scan news and social media for early signals of emerging tensions."

4. "Your plan calls for shifting resources to District 4. What do you say to the Commanders of other Districts who will see this as robbing Peter to pay Paul?"

- **Answer:** "That's a critical operational concern. The data shows District 4 is a systemic outlier. This isn't about favoring one district but about deploying resources to where the threat is most concentrated and severe to protect the *entire region*. A major crime event in District 4, like an organized auto-theft ring or a spike in gun violence, often drains resources from surrounding districts anyway. By concentrating efforts to 'put out the biggest fire,' we stabilize the entire region. This should be presented as a temporary, data-supported surge to address a critical vulnerability."

5. "How would you implement the 'Strategic Shift Bidding' in practice, considering union contracts and officer morale?"

- **Answer:** "I recognize this is as much a human challenge as a technical one. I would propose a pilot program framed as a 'Proactive Patrol Initiative.' The data shows that ensuring full staffing on weekdays reduces burnout by preventing reactive chaos on Fridays. We could use incentive-based overtime for Mondays-Thursdays, rather than forced reassignments. The key is to present the data to the officers and their representatives, showing them how this plan actually creates more predictable schedules and reduces their stress during high-chaos periods."



Behavioral & Role-Fit Questions

6. "This role involves explaining complex data to non-technical audiences, including senior officers. How do you do that?"

Answer: "I focus on the 'so what?' and avoid technical jargon. For example, instead of saying 'the XGBoost model identified a high feature importance for the temporal variable,' I would say, 'the data tells us that historical patterns from last year are the single biggest predictor of crime this month.' I use clear, intuitive visuals like the ones in this presentation—hotspot maps, simple bar charts for growth rates, and the '95% vs 54%' comparison—to make the insight immediate and undeniable."

7. "Where do you see the future of AI and Data Science in policing, and how do you fit into that?"

- **Answer:** "The future is in **predictive prevention and intelligent automation**. Beyond forecasting, AI can help in cybercrime detection, analyzing complex networks of organized crime from text reports, and optimizing patrol routes in real-time. My experience, especially with building LLM applications, positions me to help YRP not just analyze the past, but to build the tools that will give our officers a decisive advantage in the future. I'm not just a data scientist; I'm a builder of solutions that enhance public safety."

On Collaboration: "The job description mentions close collaboration with GIS, Data Engineering, and front-line decision-makers. Could you describe what a typical collaborative project looks like from initiation to deployment? What does a successful partnership look like in practice?"

On the Tech Stack Evolution: "I noticed the presentation mentioned tools like Python, SQL, and Power BI. Could you describe the current on-premise or cloud data infrastructure (e.g., Azure, AWS)? Are there any planned investments or shifts in the technology roadmap for the bureau?"

On MLOps & Model Deployment: "The purpose of this role is to transform data into strategic assets that support decision-making. Could you talk about the process for taking a model from a Jupyter Notebook to a production environment? Is there a established MLOps pipeline, or is that an area for development?"

On the AI Strategy: "York Regional Police seems committed to becoming more data-driven. Is there a formal AI or data strategy in place? How does leadership envision the role of advanced analytics, like the LLM applications I mentioned, evolving in the next few years?"